

**MINISTRY OF HIGHER EDUCATION, SCIENCE AND
INNOVATIONS OF THE REPUBLIC OF UZBEKISTAN**

**NATIONAL QUALIFICATION REQUIREMENTS
FOR
THE MASTER'S SPECIALTY
70610101 - COMPUTER SYSTEMS AND THEIR
SOFTWARE**

Tashkent -2024

DEVELOPED AND INTRODUCED BY:

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APPROVED AND PUT INTO EFFECT:

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PROPOSED:

Ministry of Higher Education, Science and Innovations of the Republic of
Uzbekistan:

These Qualification Requirements are based on the "State Educational Standard of Higher Education. General Provisions" and the "State Educational Standard of Higher Education. Classifier of Higher Education Fields and Specialties," the National and Sectoral Qualification Frameworks of the Republic of Uzbekistan, professional standards, and the proposals of personnel clients, and is an official normative-methodological document.

The right to officially publish Qualification Requirements in the territory of the Republic of Uzbekistan belongs to the Ministry of Higher Education, Science and Innovations of the Republic of Uzbekistan.

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1. GENERAL DESCRIPTION

Training of master's degree students in the specialty **70610101 - Computer Systems and Their Software** is carried out in the full-time mode of study. Instruction in all forms of education is organized on the basis of the credit-module system. The standard duration of the master's degree program is two years.

1.1 Scope of Application

1.1.1 Application of the Qualification Requirements

These Qualification Requirements define the set of requirements for all higher education institutions that provide training for master's degree students in the specialty **70610101 - Computer Systems and Their Software**.

1.1.2. Main Users of the Qualification Requirements

The main users of these Qualification Requirements for this specialty are the following:

Administrative staff of higher education institutions responsible for developing and updating qualification requirements, curricula, and academic programs for this specialty and for effectively implementing the educational process on their basis, as well as ensuring, within their authority, the required level of graduate preparedness, including rectors, vice-rectors, heads of academic departments, deans, department chairs, and academic staff;

Students of higher education institutions who complete the curriculum and academic programs of the specialty;

State Attestation Commissions responsible for assessing the level of preparedness of master's degree graduates;

Authorized state bodies responsible for educational governance;

Bodies responsible for financing higher education institutions;

Authorized state bodies responsible for accreditation and quality control in the higher education system;

Employers, personnel customers, organizations, enterprises, and other relevant stakeholders;

Bachelor's degree graduates applying for admission to higher education institutions, their parents, and other interested parties.

1.2. Description of Professional Activities

1.2.1 Areas of Professional Activity

The areas of professional activity include the design, development, administration, modelling, analysis, and research of computer systems, software systems, computer networks, distributed and parallel computing systems, information systems, and ICT infrastructure. They also include scientific research, scientific-pedagogical activity, and the development of software and hardware-software solutions for various sectors of the economy, education, industry, and public administration.

1.2.2 Objects of Professional Activity

Graduates may work as engineer-programmers in state and non-state organizations, enterprises and institutions, companies (firms), production associations, and sectoral enterprises.

Graduates of the master's specialty **70610101 - Computer Systems and Their Software** shall have the right to engage in pedagogical activities by teaching disciplines related to information and communication technologies and engineering in higher education, retraining and advanced training institutions, and professional education institutions.

- research activities and processes at the Academy of Sciences, sectoral research institutes and centers, and higher education institutions;
- mathematical, technical, and software support of information and communication technologies, computer systems and networks;
- computer-aided design systems, their hardware support, and software components of control devices;
- hardware and software used in the medical field;

- aviation;
- effective processes for the introduction, development, and operation of hardware-software tools;
- pedagogical and information technologies.

1.2.3 Types of Professional Activities

Graduates of the **70610101 - Computer Systems and Their Software**

Master's Degree Program may engage in the following types of professional activities:

- scientific research and pedagogical activities;
- design, development, and operational activities;
- organizational and managerial activities;
- experimental and research activities;
- information and analytical activities;
- production and management activities;
- engineering and entrepreneurial activities.

1.2.4 Professional Functions and Responsibilities

In accordance with **Level 7 of the National Qualifications Framework** and taking into account the areas, objects, and types of professional activity of master's graduates in the specialty **70610101 - Computer Systems and Their Software**, graduates shall be able to effectively perform the following professional functions:

In scientific research and pedagogical activities, graduates shall be able to:

- study scientific sources of scientific and technical information related to computer technologies published in the Republic and abroad;
- participate directly in carrying out scientific research in the field;
- participate in collecting, processing, analyzing, and systematizing scientific and technical data on a given topic or assignment;

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- analyze new scientific results, scientific literature, or research projects relevant to the specialty;
- apply new technologies to solve current practical problems in software development in the field of computer engineering;
- research and develop models, algorithms, and methods within the topics of ongoing research projects;
- conduct experimental research based on standard methodologies and other approaches and process the results obtained;
- prepare scientific documents and reports;
- prepare scientific reviews on the topics of ongoing research, and compile abstracts and bibliographies;
- participate in scientific seminars and scientific processes;
- prepare articles for journals corresponding to scientific topics.

In design, development, and operational activities, graduates shall be able to:

- design the information and software support of information and communication technology complexes, systems and networks, computer-aided design systems, and microprocessor-based information processing systems;
- develop and research mathematical, information, and simulation models related to the topics of experimental and practical work being carried out;
- develop and apply standard technological processes;
- prepare design and software documentation.

In organizational and managerial activities, graduates shall be able to:

- organize corporate training based on e-learning and m-learning technologies and develop corporate databases;
- develop and implement quality management processes for production activities;

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- create modern information and communication technology systems and develop methods and mechanisms for monitoring and assessing the quality of production processes related to their operation;
- participate in monitoring production processes for compliance with environmental protection and occupational safety requirements.

In experimental and research activities, graduates shall be able to:

- train students in the field of computer engineering;
- teach specialists from various fields modern technological concepts, tools, and practices through training sessions, seminars, or academic guidance;
- start and manage businesses based on digital technologies in the field of computer engineering;
- analyze market opportunities for engineering products;
- develop innovative products or services.

In information and analytical activities, graduates shall be able to:

- assess the effectiveness of projects;
- prepare reports based on the results of information and analytical activities;
- evaluate the effectiveness of managerial decisions.

In production and management activities, graduates shall be able to:

- develop proposals for improving and increasing the effectiveness of technical measures and organizational activities being adopted;
- study and summarize the experience of other institutions, organizations, and enterprises in methods of ensuring information security in order to improve and increase the effectiveness of work related to the protection of state secrets and information;

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- prevent unauthorized access to information in information and communication networks, conduct technical and economic analysis of production, and determine its technical and economic indicators;
- manage protection systems in information and communication networks in the field of management;
- understand how to improve the effectiveness of information protection using artificial intelligence methods, as well as study and summarize the experience of other institutions, organizations, and enterprises in the field of protecting state and other types of secrets.

In engineering and entrepreneurial activities, graduates shall be able to:

- develop start-up projects based on digital technologies in the field of computer engineering, and start and manage businesses;
- analyze market opportunities for engineering products;
- develop innovative products or services.

2. REQUIREMENTS FOR PROFESSIONAL COMPETENCIES

Graduates of the **70610101 — Computer Systems and Their Software Master's Degree** Program must possess the following professional competencies:

- possess a system of knowledge related to scientific and scientific-technical worldviews, know the fundamentals of general methodological disciplines, and understand current issues of state policy;
- be able to independently analyze social problems and processes;
- be able to independently acquire and improve new knowledge and organize their work on a scientific basis;
- be able to use regulatory legal documents in their activities;
- possess a culture of thinking, a specific style of reasoning, and the ability to express oral and written speech clearly;

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- be able to creatively and critically review and analyze acquired knowledge, use it in scientific and scientific-technical creative activity, and make well-founded independent decisions;
- develop social responsibility aimed at understanding the outcomes of scientific and scientific-technical creative activity;
- apply the basic laws of the disciplines mastered in professional activity, classify methods, and use methodological principles in scientific and scientific-technical creative activity;
- master one foreign language as a means of scientific communication and professional exchange;
- master methods of collecting, storing, processing, and using information;
- be able to make well-founded independent decisions in professional activity;
- be able to distinguish between information, knowledge, and data, and use information technologies;
- understand the essence and importance of information technologies in today's information society, recognize the risks and threats of information attacks, and comply with the basic requirements of information security;
- possess the main methods and tools for obtaining, storing, and processing information from the Internet, and have computer skills as a means of information management;
- effectively use information and pedagogical technologies in pedagogical activity;
- apply an innovative approach to improving the quality and effectiveness of education and scientific-technical creative activity;
- prepare project proposals for participation in projects announced in the Republic and abroad based on the results of scientific-technical creative activity;
- possess skills in preparing project proposals for participation in projects announced by state, non-state, and non-profit organizations;

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- conduct scientific and applied research, process experimental results, draw scientifically grounded conclusions based on them, prepare and edit scientific articles, organize and conduct scientific seminars, conferences, and symposia, and develop scientific projects;
- possess skills in preparing project proposals for participation in projects announced by state, non-state, and non-profit organizations;
- possess skills in organizing and managing production;
- optimize the organizational structure of organizations, enterprises, and their individual divisions, organize and manage the continuous improvement of employees' professional training levels, plan the effective use of labor potential, and establish a labor incentive system;
- apply data science for the effective use of production and labor resources in enterprises across networks and sectors;
- develop and implement mathematical, information, and simulation models related to the topics of theoretical and practical work being carried out;
- conduct research and analytical work to identify influencing factors when using data science across networks and sectors;
- conduct economic expert examination of regulatory legal documents.

3. REQUIREMENTS FOR SCIENTIFIC ACTIVITY

Scientific activity shall provide for the mastery of scientific research methodology and the performance of research and scientific-pedagogical work by students at specialized scientific and educational institutions.

Requirements for research work and preparation of the master's dissertation:

- develop practical skills for independent research activity;
- form knowledge and skills related to conducting scientific research using modern technological tools, analyzing and presenting research results, and preparing scientific articles;

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- develop students' ability to use information databases based on the latest achievements of science, engineering, and technology in the field and to apply them in preparing the master's dissertation.

Scientific and pedagogical work:

- develop pedagogical activity skills and mastery using modern pedagogical and information technologies and interactive methods of educational work;
- develop the ability and skills to organize the scientific and methodological support of the educational process.

Scientific internship (practical experience):

In the training of master's degree students, the scientific internship is conducted in the 4th semester. It shall provide opportunities to carry out research work in the field, deepen theoretical and practical knowledge, become familiar with modern equipment and technologies in science, the field, and other sectors, develop relevant practical, professional, and research competencies, and ensure effective professional adaptation. The scientific internship of a master's student assigned to such internship is expected to be completed according to an individual schedule of the educational process

4. STRUCTURE OF THE MODULE CATALOGUE

The module catalogue is structured to ensure the systematic development of advanced theoretical knowledge, professional competencies, research skills, scientific and pedagogical abilities, and practical skills in computer systems and their software. It consists of compulsory courses, elective courses, scientific and pedagogical work, qualification internship, and research work related to the preparation and defense of the Master's dissertation.

No.	Course Qualification Code	Academic Subjects, Blocks, and Activity Types	Total Workload, in Hours	Credits	Semester
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		Compulsory modules			
1.01	ILTM1106	Methodology of scientific research	180	6	1
1.02	MFOM1106	Methodology for Teaching Specialised Subjects	180	6	1
1.03	ATL1106	Design of Information Systems	180	6	1
1.04	KTMB1106	Administrative Management of Computer Networks	180	6	1
1.05	ALTQ1206	Design and Analysis of Algorithms	180	6	2
1.06	PHA1206	Foundations of Parallel Computing	180	6	2
2.00		Elective modules	720	24	2, 3
2.01.	JVT104 TQT104	Java Technologies. Distributed Systems.	120	4	2
2.02.	SINT104 BTH104	Artificial Intelligence and Neural Networks. Cloud Technologies.	120	4	2
2.03.	HUM204 VIYAT204	Computational Methods and Modelling. Web Application Development Technologies.	120	4	3
2.04.	PRI204 MSHO'204	Prompt Engineering. Machine Learning.	120	4	3
2.05.	CHJMM204 SIMA204	Mathematical Modelling of Nonlinear Processes Mathematical.	120	4	3

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		Foundations of Artificial Intelligence.			
2.06.	EKH204 PDTM204	Electronic Government. Fundamentals of the Django Framework.	120	4	3
		Qualification: Computer Systems Engineer	Engineer-Programmer, Teacher-Researcher		
3.00		Scientific Activity	1800	60	
3.01		Scientific and Pedagogical Work and Qualification Internship	600	20	1, 2, 3, 4
3.02		Research Work, Preparation and Defense of the Master's Dissertation	1200	40	1, 2, 3, 4
		GRAND TOTAL	3600	120	

Bibliographic Information

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Type of professional activity, specialty, object of professional activity, area of professional activity, core master's curriculum and academic programs, profile, learning outcomes, networks and sectors, qualification requirements, master's educational process, economy, independent learning, production, scientific research process, organizational and managerial activity, service provision, qualification internship, state attestation, academic course block, higher education institution, educational process, management process, visual design, algorithms, computing equipment, software, modeling, mathematical model, information and communication systems, modern research methods, information and modern pedagogical technologies, organization of scientific research, virtual electronic knowledge resources, didactics, theory.

Developers, Related Higher Education Institutions Agreed Upon, and Personnel Consumers

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Seal June 14, 2024

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